



LUXURY HOUSING SALES ANALYSIS BENGALURU

REAL ESTATE ANALYTICS PROJECT

PRESENTED BY : ABITHA

DOMAIN:
REAL ESTATE
BUSINESS INTELLIGENCE
DATA ANALYTICS

TOOL STACK
PYTHON | SQL | POWER BI

Project Overview

This project analyzes Bengaluru's luxury real estate market using Python, SQL, and Power BI.

The dataset includes 1,00,000+ property records covering pricing, configurations, buyer profiles, sales channels, booking status, and possession stages.

Python was used for data cleaning and feature engineering, SQL for structured storage and validation, and Power BI for interactive dashboards.

The analysis delivers insights into market trends, builder performance, booking conversion, and geographical distribution, and top-performing builders.



Business Objective

- Analyze luxury housing sales trends
- Identify top-performing builders and locations
- Understand buyer behavior and booking conversion
- Support data-driven real estate decision-making





DATASET OVERVIEW

Dataset size: 1,00,000+ records

Key Attributes captured



Builder / Developer



Micro_Market



Ticket_Price_Cr



Configuration



Booking_Status



Sales_Channel



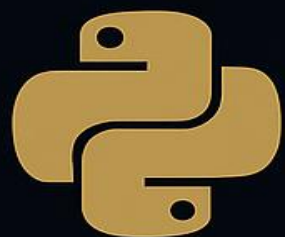
Possession_Status





PROJECT ARCHITECTURE

Analytics Workflow



Python

Python:
Data
preparation



SQL

SQL:
Storage and
validation



Power BI

Power BI:
Visualization
and insights

DATA CLEANING

Loaded raw CSV using Pandas

Standardized inconsistent formats
(price, text fields)

Handled missing and null values

Normalized categorical columns



FEATURE ENGINEERING

Created analytical features:

- Price_per_Sqft
- Quarter_Label
- Booking_Flag



Prepared clean dataset for database insertion



SQL: DATA LOADING

- Designed relational table schema
- Inserted cleaned data using Python and SQLAlchemy
- Enabled structured querying for analysis

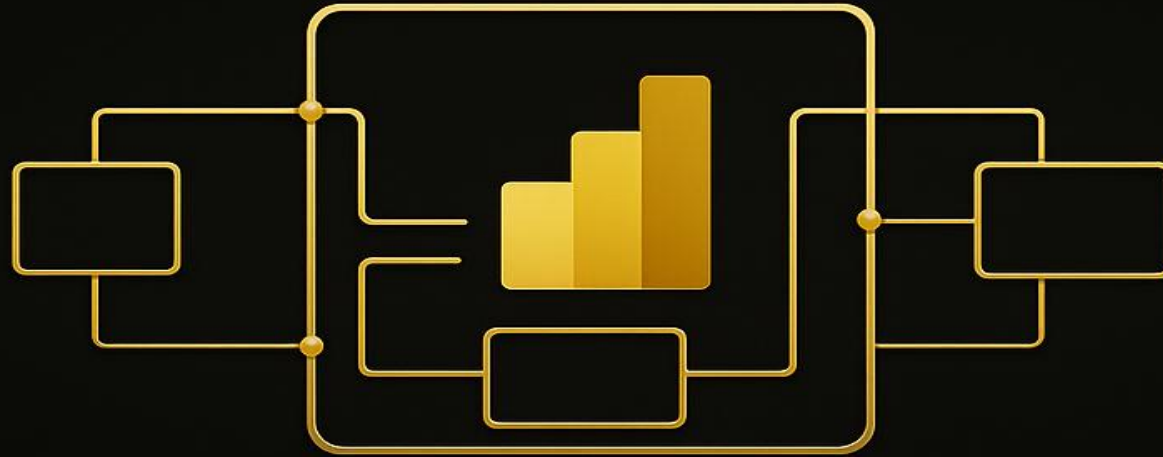
SQL: Data Validation

- ✓ Validated total record counts
- ✓ Verified revenue aggregation by builder
- ✓ Checked average amenity score across micro-markets
- ✓ Ensured data integrity and consistency



POWER BI

Data Modeling



🔗 Connected Power BI directly to SQL database

⚠ Built optimized data model and DAX measures

🔧 Enabled interactive filtering and aggregation

Market & Builder Analysis



Analyzed quarterly booking trends



Compared builders by total revenue and ticket size



Identified top-performing builders



Booking & Demand Analysis



Evaluated booking
conversion rates



Analyzed demand
by configuration
(3BHK / 4BHK / 5BHK)



Studied amenity
impact on booking
success

Luxury Housing Market – Bengaluru

Sales & Location Insights

- Compared sales channel efficiency
- Analyzed possession status influence
- Visualized geographical distribution of luxury projects



KEY BUSINESS INSIGHTS

- Certain builders dominate luxury revenue
- Specific micro-markets show higher demand
- Sales channel selection impacts conversion
- Amenities positively influence booking success



Conclusion



Demonstrated a complete end-to-end analytics pipeline



Converted raw data into business insights



Built enterprise-ready dashboards



Project showcases strong analytical and visualization skills



SCREENSHOT

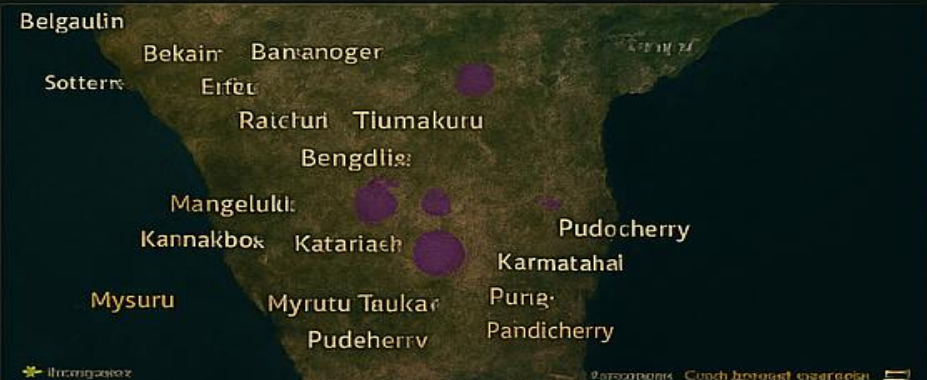


SCREENSHOT

Quarterly Builder Contribution

Developer Name	01	03	03	04
Tata Housing	292,33	20,915	202,38	302,14
Endeasy	258.99	202,15	202,58	302,34
Sobha	276,29	206,79	204,34	305,73
RMZ	283,35	202,35	282,39	302,59
Presings	283.82	299,37	202,58	302,34
Puravankara	250,78	202,52	204,31	302,38
LdTRealty	286,45	282,14	282,38	302,54
Brigade	88,40	35,54	302,35	302,34
Total Builder Revenue	288,225	270,135	302,35	1.152,33

Luxury Housing Project Concentration Across Bangalore



1,16 M

Total Revenue (₹)

107,12 K

Top Builder Revenue (₹)

★ TOP PERFORMERS

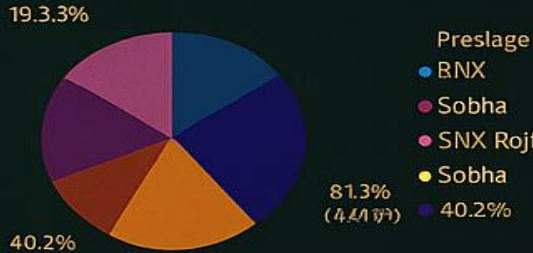
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Booking Success %

Possession Status Analysis



Leading Builders by Revenue



Top 5 Builders by Booking Success

Builder	Booking Success %
Devonar, Nurier	0.31
Indigo	0.51
pee	0.50
L4T Realty	0.50
Sobha	0.30
Total	0.51

